

Week 13: Target Trial Emulation

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Overview

Welcome to our final substantive week of the course. Over the past eleven weeks, we have amassed a powerful toolkit for causal inference: DAGs, potential outcomes, confounding control, propensity scores, instrumental variables, and methods for time-varying exposures. We have treated these as distinct topics, each with its own set of assumptions and procedures.

This week, we synthesize everything we have learned into a single, cohesive framework: **Target Trial Emulation**. This framework, powerfully articulated by Hernán & Robins (2020), provides a structured approach to designing and critiquing observational causal inference studies. The core idea is that every causal question can and should be framed as an attempt to emulate a hypothetical, perfectly-designed (but often infeasible) randomized controlled trial—the “target trial.”

By explicitly specifying the protocol of this ideal target trial, we can then systematically evaluate how our observational analysis succeeds or fails in emulating it. This process makes our assumptions transparent, our design choices deliberate, and our limitations clear. It moves us from a scattered application of statistical methods to a principled, trial-centric approach to causal inference. This week is not about learning a new statistical method, but about learning a new way to *think*.

Learning Objectives

By the end of this week’s module and workshop, you will be able to:

1. **Describe** the seven essential components of a randomized trial protocol and articulate their analogues in the design of an observational study.
2. **Formulate** a well-defined causal question by specifying the protocol of a hypothetical “target trial” that would, if conducted, answer it.
3. **Critique** an observational study by comparing its design and analysis to a specified target trial, identifying potential sources of bias arising from deviations.
4. **Apply** the target trial framework to your own research project, clarifying your study’s eligibility criteria, treatment strategies, assignment mechanism, and causal estimand.

5. **Synthesize** the range of causal methods learned in this course as tools to be used in the “Analysis Plan” component of an emulated trial.

R Packages

This week is primarily conceptual, but we will use `ggdag` to visualize the core challenge that observational studies must overcome.

```
if (!require("pacman")) {  
  install.packages("pacman")  
}  
pacman::p_load(  
  tidyverse, # For data manipulation  
  ggdag, # For plotting DAGs  
  knitr # For table formatting  
)  
  
# A consistent plot theme  
theme_set(theme_dag_blank())
```

1. Comprehensive Topic Explanation

The Unifying Idea: Observational Studies as Attempts to Emulate a Trial

The central premise of the target trial framework is that for any causal question, there is an ideal (though perhaps hypothetical) randomized experiment that could answer it. This ideal experiment is the **target trial**. When we conduct an observational study, what we are really trying to do is use observational data and sophisticated analytical methods to emulate that target trial as closely as possible.

Why is this framework so powerful?

- **It provides a common language:** Trialists, epidemiologists, and data scientists can all understand the components of a trial protocol.
- **It forces clarity:** Vague causal questions like “What is the effect of statins?” must be refined into specific, answerable questions by defining the components of the trial.
- **It illuminates sources of bias:** Biases in observational studies can be elegantly reframed as specific deviations from the target trial’s protocol.

The Seven Components of a Target Trial Protocol

To emulate a trial, we must first explicitly design it. A trial protocol consists of seven key components. For each component, we must consider its ideal form and its observational analogue.

1. Eligibility Criteria:

- **Target Trial:** Who would be included in the trial? This involves defining the study population based on factors like age, disease status, and prior medical history. Crucially, eligibility must be determined *before* treatment assignment.
- **Observational Analogue:** This corresponds to the inclusion/exclusion criteria for our observational cohort. A common mistake is to select individuals based on information recorded after their treatment has already started, which can introduce selection bias.

2. Treatment Strategies:

- **Target Trial:** What, exactly, are the interventions being compared? A “treatment strategy” is a rule that specifies the intervention over time. It must be well-defined and realistic. “Take statins” is vague. “Initiate 40mg of atorvastatin daily at baseline and continue unless a severe side effect occurs” is a well-defined strategy.
- **Observational Analogue:** We must identify groups in our data that appear to follow the specified strategies. This can be challenging. For example, in the real world, people’s adherence to a medication plan is not perfect.

3. Assignment Procedure:

- **Target Trial:** How are individuals assigned to a treatment strategy? In an ideal trial, this is **random assignment**. Randomization is the key to ensuring that, on average, the treatment groups are exchangeable at baseline.
- **Observational Analogue:** This is the most critical point of departure. In observational studies, assignment is **not random**. People’s doctors and they themselves choose treatments based on prognostic factors. This is the origin of **confounding**. Our causal inference methods (propensity scores, standardization, IV) are all attempts to adjust for the lack of randomization.

4. Follow-up Period:

- **Target Trial:** When does follow-up begin and end for each individual? Follow-up must begin at the moment of randomization (Time Zero).
- **Observational Analogue:** We must define a “Time Zero” in our observational data that corresponds to the start of the treatment strategy. Forgetting to do this is a common source of “immortal time bias,” where, for example, a person must survive for a period of time before they can be classified as “treated,” leading to a spurious protective effect.

5. Outcome:

- **Target Trial:** What is the outcome of interest, and how and when is it measured? (e.g., incidence of myocardial infarction within 5 years).
- **Observational Analogue:** The outcome definition should be identical. We must ensure it is ascertained equally in all treatment groups to avoid detection bias.

6. Causal Contrast (Estimand):

- **Target Trial:** What is the specific causal question? Are we interested in the **Intention-to-Treat (ITT) effect** (the effect of being *assigned* to a strategy, regardless of adherence) or the **Per-Protocol effect** (the effect of actually *adhering* to the strategy)?
- **Observational Analogue:** This choice has major implications for our analysis. Emulating an ITT effect is often more straightforward (comparing groups based on initial treatment), while emulating a per-protocol effect requires methods like MSMs to handle time-varying confounding and adherence.

7. Analysis Plan:

- **Target Trial:** How will the causal contrast be estimated? In a simple trial, this might be a comparison of risks or means.
- **Observational Analogue:** This is where our entire toolkit comes into play. The analysis plan specifies the method chosen to adjust for the lack of randomization—be it IPTW to handle time-varying confounding for a per-protocol analysis, or a baseline propensity score model to emulate a simple ITT analysis.

2. Intuitive Clarifications of Challenging Ideas

Analogy: The “Coffee and Heart Disease” Study

Imagine we want to answer the question: “Does drinking coffee cause heart disease?” This is a vague question. Let’s use the target trial framework to make it concrete.

Designing the Target Trial:

- **Eligibility:** We might enroll adults aged 40-65 with no prior history of heart disease.
- **Treatment Strategies:** We’ll compare two strategies: (1) “Abstain from all caffeinated coffee for 10 years” vs. (2) “Drink 2 cups of caffeinated coffee every morning for 10 years.”
- **Assignment:** We would randomly assign eligible participants to one of the two strategies.
- **Follow-up:** We would start follow-up at the moment of randomization and continue for 10 years.
- **Outcome:** The outcome is the incidence of a major cardiovascular event (e.g., heart attack, stroke).

- **Causal Contrast:** We might be interested in the per-protocol effect—the effect among those who actually stuck to their assigned coffee-drinking (or abstaining) plan.
- **Analysis Plan:** Compare the 10-year risk of cardiovascular events between the two groups.

Emulating the Trial with Observational Data: Now, we take our observational dataset (e.g., the NHANES survey) and try to emulate this trial.

- **Eligibility:** We select people in our data who meet the eligibility criteria *at the start of our observation period*.
- **Treatment Strategies:** We find people in the data whose reported coffee intake roughly matches our two strategies.
- **Assignment:** This is the problem. In the real world, people who drink coffee are different from those who don't (they might smoke more, exercise less, etc.). This is **confounding**.
- **Analysis Plan to Fix Assignment:** Our analysis plan must now include a method to adjust for these differences. Since behaviors like smoking can change over time and influence future coffee drinking, we likely have time-varying confounding. Therefore, our analysis plan would specify using a Marginal Structural Model with IPTW to estimate the per-protocol effect we desire.

By structuring our thinking this way, we see that the choice of using an MSM was not arbitrary; it was dictated by the specific causal question (per-protocol effect) and the specific challenges (time-varying confounding) that arose from emulating our target trial.

Intention-to-Treat vs. Per-Protocol: A Key Distinction

Causal Question	Intention-to-Treat (ITT) Effect	Per-Protocol Effect
What it answers	What is the effect of <i>prescribing</i> or <i>assigning</i> a treatment strategy?	What is the effect of <i>adhering</i> to a treatment strategy?
Analogy to Trial	Compares groups based on their randomized assignment, regardless of what they actually did (“as randomized”).	Compares groups based on what treatment they actually received and adhered to.
Bias in an Ideal Trial	Unbiased for the effect of assignment. Preserves the benefits of randomization.	Can be biased even in a trial, as adherence is a post-randomization behavior.
Observational Emulation	Simpler. Requires adjusting for baseline confounders only. Example: Propensity score matching on initial treatment.	More complex. Requires adjusting for baseline <i>and</i> time-varying confounders that affect adherence. Example: MSMs.

Causal		
Question	Intention-to-Treat (ITT) Effect	Per-Protocol Effect
Usefulness	Pragmatic. Useful for policy makers deciding whether to recommend a strategy.	Explanatory. Useful for scientists understanding the biological effects of a treatment if taken as intended.

3. Simple, Hand-Calculation Numerical Example(s)

Since this week is conceptual, we won't do a numerical calculation of an effect. Instead, our "by-hand" example will be to **design a target trial protocol on paper**.

Causal Question: Does starting a vegan diet cause weight change in overweight adults?

Step-by-Step Protocol Design:

1. Eligibility Criteria:

- Include adults aged 18-65.
- Include individuals with a Body Mass Index (BMI) between 25 and 35 kg/m².
- Exclude individuals who are already vegetarian or vegan.
- Exclude pregnant individuals.
- Exclude individuals with medical conditions that would contraindicate a vegan diet.

2. Treatment Strategies:

- **Strategy 1 (Intervention):** Follow a vegan diet for 12 months. A vegan diet is defined as excluding all meat, poultry, fish, dairy, eggs, and honey.
- **Strategy 2 (Control):** Continue with your usual diet for 12 months.

3. Assignment Procedure:

- **Target Trial:** Randomly assign eligible participants 1:1 to either Strategy 1 or Strategy 2.

4. Follow-up Period:

- Follow-up begins at the time of randomization (Day 0).
- Follow-up ends after 12 months.

5. Outcome:

- The primary outcome is the change in body weight (in kg) from baseline to 12 months.

6. Causal Contrast (Estimand):

- We are interested in the **Intention-to-Treat effect**: the effect of being assigned to the vegan diet strategy, regardless of how well participants adhered to it. This answers the pragmatic question, “What is the effect of recommending a vegan diet?”

7. Analysis Plan:

- **Target Trial:** At 12 months, calculate the mean weight change in the group assigned to the vegan diet and the mean weight change in the group assigned to the usual diet. The ATE is the difference between these two means.
- **Observational Emulation:** If we were using observational data, our analysis plan would need to adjust for baseline confounders (e.g., age, sex, baseline activity level) that differ between people who spontaneously adopt a vegan diet and those who don’t. We could use propensity score matching or g-computation on the baseline covariates.

This simple exercise transforms a vague question into a precise, answerable scientific query.

4. R Code Examples & Interpretation

The key challenge in observational studies is the non-random assignment. We can visualize this difference with ggdag.

The Ideal Trial vs. The Observational Reality

```
# In an ideal trial, assignment (A) is random, so it has no parents.
trial_dag <- dagify(
  Y ~ A,
  exposure = "A",
  outcome = "Y"
)
ggdag(trial_dag) + labs(title = "Ideal Randomized Trial")

# In an observational study, confounders (L) affect both assignment (A) and outcome (Y).
obs_dag <- dagify(
  Y ~ A + L,
  A ~ L,
  exposure = "A",
  outcome = "Y"
)
```

```
ggdag(obs_dag) + labs(title = "Observational Reality")
```

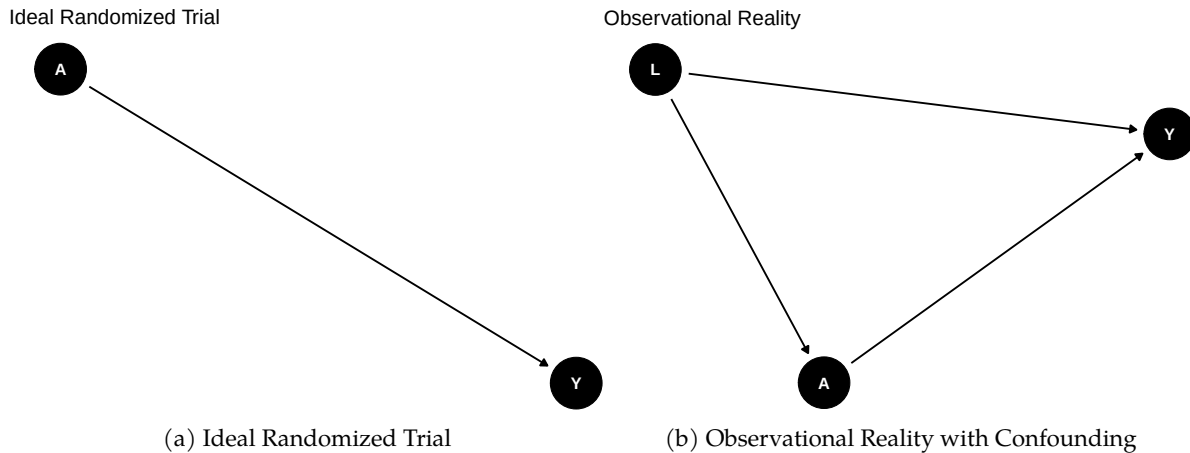


Figure 1: Causal DAGs for a Randomized Trial vs. an Observational Study

Interpretation:

- **Ideal Trial DAG:** The beauty of a randomized trial is its simplicity. There are no arrows pointing into the exposure A. This means there are no backdoor paths between A and the outcome Y. The association between A and Y is the causal effect.
- **Observational DAG:** This is the world we usually live in. There is a confounder L that is a common cause of both A and Y, creating a backdoor path ($A \leftarrow L \rightarrow Y$). If we naively calculate the association between A and Y, we get a mix of the true causal effect and spurious association from this open backdoor.

The target trial framework tells us that the goal of our observational analysis plan is to find a way to make the observational DAG *behave like* the trial DAG. Methods like propensity score weighting or g-computation are tools designed to close that backdoor path, allowing us to isolate the true causal arrow $A \rightarrow Y$.

5. Hands-On R Lab

Lab: Designing a Target Trial for Your Final Project

Purpose: This lab is a capstone workshop. It is designed to help you synthesize the concepts from the course and apply them to your own research question using the structured thinking of the target trial framework.

Lab Objectives:

1. Take a real-world causal question and formalize it by specifying the seven components of a target trial protocol.
2. Identify the key points where an observational study attempting to answer this question would deviate from the ideal trial.
3. Choose an appropriate analytical method from this course to address those deviations.

Instructions: For this lab, you will not be writing R code to analyze data. You will be writing a research protocol in a Quarto document. Choose a causal question you are interested in (this could be for your final project or another topic of interest). Create a new Quarto document and, using the seven components below as your section headers, write out the protocol for the ideal target trial that would answer your question.

Then, for each section, add a subsection called “**Observational Emulation Challenges**” where you describe the specific problems you would face trying to implement this part of the protocol using real-world observational data.

Example Project: Effect of Statin Therapy on Heart Disease

Here is a filled-out example to guide you.

1. Eligibility Criteria

- **Target Trial Protocol:** We will enroll individuals aged 40-75 who have not had a prior cardiovascular event but have at least one risk factor (e.g., hypertension, high cholesterol).
- **Observational Emulation Challenges:** In our electronic health record data, we must be careful to define eligibility based only on information available *before* the first statin prescription (or a corresponding index date for non-users). If we include people who had a heart attack *before* starting statins, we will have selection bias.

2. Treatment Strategies

- **Target Trial Protocol:** We will compare two strategies: (1) “Initiate a moderate-intensity statin (e.g., atorvastatin 20mg) at baseline” vs. (2) “No statin therapy.”
- **Observational Emulation Challenges:** In the real world, people may not adhere to their prescription, or they may switch doses or types of statins. We must decide if we are emulating an “intention-to-treat” effect (based on first prescription) or a “per-protocol” effect, which would require tracking adherence over time.

3. Assignment Procedure

- **Target Trial Protocol:** Randomly assign eligible participants to one of the two strategies.

- **Observational Emulation Challenges:** This is the main deviation. In our data, doctors prescribe statins to patients they believe are at higher risk. This is **confounding by indication**. The groups are not exchangeable at baseline.

4. Follow-up Period

- **Target Trial Protocol:** Follow-up begins at randomization and continues for 5 years.
- **Observational Emulation Challenges:** We must define Time Zero carefully. For treated individuals, it's the date of their first prescription. For untreated individuals, we must assign a comparable index date. If we don't, we can introduce **immortal time bias**. We also have to worry about loss to follow-up (censoring).

5. Outcome

- **Target Trial Protocol:** The primary outcome is the first occurrence of a major adverse cardiovascular event (MACE).
- **Observational Emulation Challenges:** We must ensure that MACE is recorded with equal accuracy for both statin users and non-users. If statin users have more frequent doctor visits, events might be detected more often, leading to ascertainment bias.

6. Causal Contrast

- **Target Trial Protocol:** We will estimate the **Intention-to-Treat effect**.
- **Observational Emulation Challenges:** Emulating the ITT effect is feasible. We will compare the group that received an initial statin prescription to the group that did not.

7. Analysis Plan

- **Target Trial Protocol:** Compare the 5-year risk of MACE between the two randomized groups.
- **Observational Emulation Challenges:** To address the confounding from the non-random assignment, our analysis plan must include an adjustment method. To estimate the ITT effect, we can use **propensity score matching** based on a rich set of baseline covariates (age, sex, comorbidities, baseline cholesterol, etc.) measured at Time Zero. The goal is to create two groups that, at baseline, have a similar probability of having been prescribed a statin, thus emulating randomization.

Your Turn: Protocol Design Worksheet

Now, create a new document and fill this out for your own research question.

My Causal Question: _____

1. Eligibility Criteria

- **Target Trial Protocol:** ...

- **Observational Emulation Challenges: ...**

2. Treatment Strategies

- **Target Trial Protocol: ...**
- **Observational Emulation Challenges: ...**

3. Assignment Procedure

- **Target Trial Protocol: ...**
- **Observational Emulation Challenges: ...**

4. Follow-up Period

- **Target Trial Protocol: ...**
- **Observational Emulation Challenges: ...**

5. Outcome

- **Target Trial Protocol: ...**
- **Observational Emulation Challenges: ...**

6. Causal Contrast

- **Target Trial Protocol: ...**
- **Observational Emulation Challenges: ...**

7. Analysis Plan

- **Target Trial Protocol: ...**
- **Observational Emulation Challenges: ...**

6. Common Pitfalls & Checks

Immortal Time Bias

A frequent and subtle error in emulating trials is the incorrect handling of time. “Immortal time” is a period of follow-up during which, by design, the outcome of interest cannot occur. For example, if you define your “treated” group as “patients who took a drug for at least 6 months,” then by definition, everyone in your treated group had to survive for at least 6 months to be included. This creates a spurious survival advantage for the treated group. The solution is to always start follow-up at a consistently defined Time Zero for everyone, before treatment has begun.

! Specificity is Your Best Defense Against Bias

The target trial framework's greatest strength is that it forces you to be relentlessly specific. Vague terms like "effect," "exposure," and "outcome" are not allowed. You must define the precise strategies, population, and timeline. This specificity is what allows you to clearly see the potential mismatches between the ideal trial and your observational data, which are the sources of bias.

7. Advanced Teaching Activities & Interactivity

1. "Critique the Paper" Exercise:

- **Activity:** I will provide the abstract and methods section of a published observational study. In small groups, your task will be to reverse-engineer the target trial that the authors were implicitly trying to emulate.
- **Task:** Write down the 7 protocol components for their target trial. Then, identify the most significant deviation in the authors' observational study and discuss whether their chosen analysis plan (e.g., a simple multivariable regression) was sufficient to address it.

2. Think-Pair-Share: The "Pragmatic" Trial:

- **Think:** "Many modern RCTs are 'pragmatic trials,' where treatment assignment is random, but adherence and follow-up are left to be as they would in the real world. How does this blur the line between a trial and an observational study? Which biases that we've discussed for observational studies might still be present in a pragmatic trial, especially when estimating a per-protocol effect?"
- **Pair:** Discuss with a neighbor. Key points might include confounding by indication when analyzing adherence.
- **Share:** We'll discuss how the target trial framework helps us understand that even some RCTs are not "perfect," and the same causal principles apply.

8. Appendix: Advanced Theory & Derivations

Synthesis of Causal Methods within the Target Trial Framework

This framework allows us to see our entire course as a set of solutions to problems that arise when trying to emulate a target trial.

Problem (Deviation from Trial)	Causal Concept	Analytical Solution(s)	Week Covered
Assignment is not random. (Baseline prognostic factors differ between groups).	Confounding	Regression, Stratification, Propensity Score Matching, G-Computation	3, 4, 5, 7
Assignment is not random and we have a valid external factor that influences it.	Instrumental Variable	Two-Stage Least Squares (2SLS)	10
We want the per-protocol effect , but adherence is affected by time-varying prognostic factors.	Time-Varying Confounding	Inverse Probability Weighting (IPTW) for Marginal Structural Models (MSMs)	6
Some individuals are lost to follow-up, and this loss is related to treatment and outcome.	Selection Bias (Collider-Stratification)	Inverse Probability of Censoring Weighting (IPCW)	4, 11
The treatment has a “sharp” threshold for eligibility based on a continuous variable.	Regression Discontinuity	Local Linear Regression at the cutoff	9

The target trial framework provides the structure to identify the problem, and the methods we’ve learned throughout this course provide the tools to solve it. This structured approach is the hallmark of modern, rigorous causal inference.

References

Hernán, M. Á., & Robins, J. M. (2020). *Causal inference: What if*. Chapman & Hall/CRC.